

Haoyu Zhang

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Education

- M.S. Southwest University, Operations Research – Chongqing** Sept 2025 – present
- (Ranked 3/65): Optimization, Variational Analysis, etc.
- B.S. Donghua University, Mathematics and Applied Mathematics – Shanghai** Sept 2021 – June 2025
- (Ranked 8/61): Mathematical Modeling, Discrete Mathematics, Data Structures and Algorithms, etc.

Knowledge and Skills

Expertise: Distributed Optimization, Stochastic Optimization, Primal-Dual Algorithms

Languages: Mandarin (Native), English (CET-6), Japanese (JLPT-N2)

Programming: Python, MATLAB

Work Experience

- Math AI Training (AIGC) Intern, Appen Ltd. – Chongqing, China** Jan 2025 – Apr 2025
- Contributed to the development of mathematical vertical capabilities for Large Language Models (LLMs) to enhance their reasoning performance in specialized domains.
 - Engineered complex reasoning solution chains and step-by-step logical paths using LaTeX and Markdown, facilitating the batch production of high-quality training data.

Research Experience

- Multi-Species Ecological Competition & Regression Analysis System** Dec 2022 – May 2023
- Modeler of research group
- Developed a regression analysis system to simulate complex interspecies competitive relationships based on the Lotka-Volterra model.
- Fluid Dynamics & Thermal Coupling Modeling for Greenhouse Environments** Nov 2023 – Apr 2024
- Leader of research group
- Constructed a coupling model for greenhouse gas concentrations and temperature distributions utilizing Navier-Stokes and heat conduction equations.

Selected Honors & Awards

- First-Prize Graduate Academic Scholarship, Southwest University** 2025
- Tianji Annual Scholarship, Donghua University** 2024
- Honorable Mention, Mathematical Contest in Modeling (MCM)** 2023
- Third-Prize Undergraduate Academic Scholarship, Donghua University** 2022 & 2023
- Shanghai Municipal Prize, Contemporary Undergraduate Mathematical Contest in Modeling (CUMCM)** 2023
- Third Prize, Asia-Pacific Mathematical Contest in Modeling (APMCM)** 2023